

Final Project Option A — Simulation Optimization Extension

Module M11 | Chapter 13 | Tier: Advanced | Extension worth 35 of 100 final project points

Module: M11 Chapter: 13 Tier: Advanced Included in final project submission

Overview

This option extends the final project by asking you to move beyond scenario comparison and into optimization over a design space. It corresponds to **Option A** in `assignments/final_project.tex`.

Choose one simulation model from your project and define a nontrivial decision vector to optimize. The decision may be staffing, capacity, routing thresholds, reorder points, or another policy variable approved by the instructor.

Minimum standard. A strong submission will make it clear that optimization is not just picking the smallest noisy sample mean. The report must separate exploratory search from final validation, and every recommendation must be accompanied by a confidence interval.

1 Required Components (35 total pts)

1.1 1. Optimization Problem Statement (8 pts)

Formally define:

- **Decision variables $\mathbf{x}$:** the parameters under your control (e.g., number of servers c , reorder point s , lane threshold τ). State units and types (integer, real, binary).
- **Feasible set $\mathcal{X}$:** explicit bounds and any integrality or combinatorial constraints.
- **Objective function $f(\mathbf{x})$:** the performance measure to minimize (or maximize), including units.
- **At least one operational constraint:** e.g., “mean wait ≤ 5 minutes” or “staffing cost $\leq \$5,000/\text{day}$ ”.

1.2 2. Search Strategy (10 pts)

Use one of the following methods (or another instructor-approved gradient-free approach):

- **Coarse grid search with refinement:** enumerate a coarse grid, identify the promising region, then refine.
- **Local search:** start from an initial point and iteratively improve using a stated neighborhood definition.

- **Response-surface or surrogate-guided search:** fit a surrogate model to evaluated points and use it to guide the next evaluation.

For the chosen method, report:

- the search algorithm and its parameters (step size, neighborhood radius, etc.),
- the total number of simulation runs used in the search phase, and
- how you handled noise (i.e., how many replications per candidate design).

1.3 3. Replication Logic (7 pts)

- State how many replications were used during the pilot (search) phase and justify the choice.
- Explain how the final candidate was re-evaluated (it must use more replications than the pilot phase).
- Report a **95% confidence interval** for the objective at the recommended design.

1.4 4. Results (7 pts)

Present:

1. A **table** of all evaluated candidate designs with their estimated objective values and standard errors.
2. **One plot** showing the search trajectory (objective vs. iteration number), a surrogate surface, or a response surface slice.
3. A **final recommendation**: the recommended design $\mathbf{x}^*$, its estimated objective with CI, and how it compares to the baseline from the core study.

1.5 5. Limitations (3 pts)

- Identify at least **two limitations** of your search (e.g., local vs. global optimality, budget constraints, constraint handling).
- State whether the recommended design is robust to plausible changes in the arrival rate or service distribution.

2 Deliverable

Add a section titled **Simulation Optimization Extension** to your final project report, following this structure:

1. Problem formulation
2. Search method and experimental budget
3. Evaluated candidate designs (table)
4. Search trajectory or surrogate surface (figure)

5. Recommendation with CI and comparison to baseline

6. Limitations

Include the optimization driver script or notebook in the code repository. The script must be runnable from the repo root after `pip install -e simdes/` and must reproduce the results table.

Grading (within the 35-point extension allocation)

Component	Points
Optimization problem statement (well-defined)	8
Search strategy (described and justified)	10
Replication logic (pilot vs. final, CI reported)	7
Results (table + figure + recommendation with CI)	7
Limitations (at least two, specific)	3
Total	35

Full rubric: [rubrics/final_project_rubric.pdf](#) | Exemplar reports: the instructor package (available to instructors on request).